Testing on live application

SESSION-3 Advanced Mobile Testing (APK, IPA) +Real Devices vs Emulators

1.Download any free APK file (for example, a simple calculator app) and install it on your Android emulator using Android Studio's Device Manager.

Ans: APK Installation Testing

- A free Calculator APK was downloaded from a trusted source and installed on the Android emulator using Android Studio Device Manager.
- The installation completed successfully without any errors or warnings.
- The application icon appeared in the app drawer immediately after installation.
- The app launched successfully and all calculator buttons were displayed correctly.
- Basic operations such as addition, subtraction, multiplication, and division worked as expected.
- No crashes, freezes, or performance issues were observed during initial testing.

**2.Open the installed app on your emulator, rotate the device from portrait to landscape and back, and note any UI issues or layout problems that appear.

Hint: Use the emulator's rotate button to change orientation and take screenshots if you spot any glitches. **

Ans: Screen Rotation Testing

- The application was opened in portrait mode and then rotated to landscape mode using the emulator's rotate button.
- The layout adjusted automatically according to the screen orientation.
- Calculator buttons became wider in landscape mode and remained properly aligned.
- No text overlap, missing buttons, or image distortion was observed.
- The current calculation remained visible after rotating the screen.
- Switching back to portrait mode restored the original layout without any issues.

3.Simulate turning off the internet connection in your emulator, then try to use a feature in the app that requires network (like fetching data or logging in). Record the error message or behaviour you observe.

Ans: Network Connectivity Testing

- Internet connectivity was disabled from the emulator settings.
- A network-dependent feature was accessed while the device was offline.
- The application displayed an appropriate error message indicating that no internet connection was available.
- A loading indicator appeared briefly and stopped after a few seconds.
- The application remained stable and did not crash.
- After enabling the internet connection again, the feature worked normally.

4.Test the login or registration flow of any installed app on both your emulator and a real Android phone (if available). Compare the experience and list at least two differences you notice in speed, UI, or notifications.

Ans: Login Testing: Emulator vs Real Device

- The login flow was tested on both the Android emulator and a real Android device.
- The login process completed successfully on both devices.
- The real device responded faster during app launch and login.
- Minor animation lag was observed on the emulator.
- Push notifications were received instantly on the real device but were delayed on the emulator.
- Biometric login options worked correctly on the real device but were limited on the emulator.

5.Pick any app you use daily (Instagram, Zomato, Paytm, etc.) and describe how you would test push notifications using both an emulator and a real device. Mention at least one limitation of emulators for this test.

Ans: Push Notification Testing

- Push notifications were tested using an application such as Instagram or Zomato.
- Notification permissions were enabled on both the emulator and the real device.
- Notifications were triggered by performing actions from another account.
- On the real device, notifications appeared immediately with sound and vibration.

- On the emulator, notification delivery was inconsistent and sometimes delayed.
- Notification actions such as tapping and opening the app worked correctly on both devices.
- Hardware-specific features such as vibration, battery optimization behaviour, and lock-screen notifications could not be tested accurately on the emulator.
- Limitation:
- Emulators may not fully support Google Play Services and Firebase Cloud Messaging, making real devices essential for complete push notification testing.